

PES UNIVERSITY

Department of Computer Science and Engineering (AIML)

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Student Club Event Ticketing & Budget Portal

Name: Ramneek Gautam | SRN: PES1UG24AM220

Architecture Selection

We chose Layered Architecture for the Student Club Event Ticketing & Budget Portal.

Architectural Choice

The portal is a centralized web-based system with distinct responsibilities for event management, budget processing, approval workflow, QR ticketing and validation, reporting, and persistent data. Layered architecture provides a clear separation of concerns and fits the structured workflow already defined for the system.

Two Reasons

1. Clear separation of responsibilities: The user interface, application services, ticket validation, reporting, and database access can be maintained as separate logical layers. This makes the system easier to understand, test, maintain, and extend.
2. Suitable for the portal's workflow: Event proposals and budgets must pass through an ordered approval process before later activities such as ticketing and venue check-in. A layered structure supports this predictable flow while keeping shared data in a centralized database.

Security Advantage

Security-sensitive operations can be isolated in the application and validation layers. QR tickets can be checked for authenticity and usage status before entry is granted, while database access remains behind controlled application services. This reduces direct exposure of stored event, attendee, budget, and approval data.

Performance Benefit

Frequently used operations such as event retrieval, QR ticket validation, and reporting can be handled by dedicated application components while the database is accessed through controlled interfaces. This reduces unnecessary coupling and allows performance-sensitive operations such as ticket validation to remain lightweight during event check-in.

Component Identification

The component model contains six major components: Web / Mobile Interface, Event & Budget Manager, Approval Workflow, QR Ticketing & Validation, Reporting & Analytics, and Portal Database. The interfaces represent event APIs, approval requests, ticket access, ticket data, validation status, report data, and database access.

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UML Component Diagram

